Unit : 1 PROGRAMS

- Write a program of performe CRUD operation using Statement interface using switch case.

```
import java.sql.*;
import java.util.Scanner;
public class EmployeeCRUD {
   public static void main(String args[]) {
      Scanner sc = new Scanner(System.in);
      try {
         // Load JDBC Driver
         Class.forName("com.mysql.cj.jdbc.Driver");
         // Create Connection
         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/test",
               "root",
               "");
         Statement st = con.createStatement();
         int choice;
         do {
            System.out.println("\n===== EMPLOYEE CRUD MENU =====");
            System.out.println("1. Insert Employee");
            System.out.println("2. Display All Employees");
            System.out.println("3. Update Employee Salary");
            System.out.println("4. Delete Employee");
            System.out.println("5. Exit");
            System.out.print("Enter Your Choice : ");
            choice = sc.nextInt();
```

```
switch(choice) {

   case 1:

      sc.nextLine();
      System.out.print("Enter Employee No : ");
      int empno = sc.nextInt();
      sc.nextLine();
      System.out.print("Enter Employee Name : ");
      String empnm = sc.nextLine();
      System.out.print("Enter Designation : ");
      String designation = sc.nextLine();
      System.out.print("Enter City : ");
      String city = sc.nextLine();
      System.out.print("Enter Salary : ");
      double salary = sc.nextDouble();
      sc.nextLine();
      System.out.print("Enter Department : ");
      String department = sc.nextLine();

      String insert = "INSERT INTO emp VALUES("
            + empno + ",'"
            + empnm + "','"
            + designation + "','"
            + city + "',"
            + salary + ",'"
            + department + "')";

      int i = st.executeUpdate(insert);
```

```
    if(i > 0)
       System.out.println("Record Inserted Successfully.");
    else
       System.out.println("Record Not Inserted.");
    break;

case 2:
    ResultSet rs = st.executeQuery("SELECT * FROM emp");
    while(rs.next()) {
       System.out.println(
             rs.getInt("empno") + "\t" +
             rs.getString("empnm") + "\t" +
             rs.getString("designation") + "\t" +
             rs.getString("city") + "\t" +
             rs.getDouble("salary") + "\t" +
             rs.getString("department"));
    }
    break;
case 3:
    System.out.print("Enter Employee No : ");
    empno = sc.nextInt();

    System.out.print("Enter New Salary : ");
    salary = sc.nextDouble();
    String update = "UPDATE emp SET salary="
          + salary + " WHERE empno=" + empno;

    i = st.executeUpdate(update);
```

```
            if(i > 0)
               System.out.println("Record Updated Successfully.");
            else
               System.out.println("Employee Not Found.");
            break;

         case 4:
            System.out.print("Enter Employee No : ");
            empno = sc.nextInt();
            String delete = "DELETE FROM emp WHERE empno=" + empno;
            i = st.executeUpdate(delete);
            if(i > 0)
               System.out.println("Record Deleted Successfully.");
            else
               System.out.println("Employee Not Found.");
            break;

         case 5:
            System.out.println("Program Terminated.");
            break;

         default:
            System.out.println("Invalid Choice.");
      }
   } while(choice != 5);
   con.close();
} catch(Exception e) {
   System.out.println(e);
}
```

```
    }
}
```

- Write a program of performe CRUD operation using PreparedStatement interface using switch case.

```
import java.sql.*;
import java.util.Scanner;
public class EmployeeCRUDPrepared {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");
            // Create Connection
            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/test",
                    "root",
                    "");
            int choice;
            do {
                System.out.println("\n===== EMPLOYEE CRUD MENU =====");
                System.out.println("1. Insert Employee");
                System.out.println("2. Display All Employees");
                System.out.println("3. Update Employee Salary");
                System.out.println("4. Delete Employee");
                System.out.println("5. Exit");
                System.out.print("Enter Your Choice : ");

                choice = sc.nextInt();
                switch(choice) {
```

```
// INSERT
case 1:
   String insert = "INSERT INTO emp VALUES(?,?,?,?,?,?)";
   PreparedStatement ps1 = con.prepareStatement(insert);
   System.out.print("Enter Employee No : ");
   ps1.setInt(1, sc.nextInt());
   sc.nextLine();
   System.out.print("Enter Employee Name : ");
   ps1.setString(2, sc.nextLine());
   System.out.print("Enter Designation : ");
   ps1.setString(3, sc.nextLine());
   System.out.print("Enter City : ");
   ps1.setString(4, sc.nextLine());
   System.out.print("Enter Salary : ");
   ps1.setDouble(5, sc.nextDouble());
   sc.nextLine();
   System.out.print("Enter Department : ");
   ps1.setString(6, sc.nextLine());
   int i = ps1.executeUpdate();
   if(i > 0)
      System.out.println("Record Inserted Successfully.");
   else
      System.out.println("Record Not Inserted.");
   break;

// DISPLAY
case 2:
   String select = "SELECT * FROM emp";
```

```
    PreparedStatement ps2 = con.prepareStatement(select);
    ResultSet rs = ps2.executeQuery();
    while(rs.next()) {
        System.out.println(
            rs.getInt("empno") + "\t" +
            rs.getString("empnm") + "\t" +
            rs.getString("designation") + "\t" +
            rs.getString("city") + "\t" +
            rs.getDouble("salary") + "\t" +
            rs.getString("department"));
    }
    break;

// UPDATE
case 3:
    String update = "UPDATE emp SET salary=? WHERE empno=?";
    PreparedStatement ps3 = con.prepareStatement(update);
    System.out.print("Enter Employee No : ");
    int empno = sc.nextInt();
    System.out.print("Enter New Salary : ");
    double salary = sc.nextDouble();
    ps3.setDouble(1, salary);
    ps3.setInt(2, empno);
    i = ps3.executeUpdate();
    if(i > 0)
        System.out.println("Record Updated Successfully.");
    else
        System.out.println("Employee Not Found.");
    break;
```

```
            // DELETE
            case 4:
               String delete = "DELETE FROM emp WHERE empno=?";
               PreparedStatement ps4 = con.prepareStatement(delete);
               System.out.print("Enter Employee No : ");
               empno = sc.nextInt();
               ps4.setInt(1, empno);
               i = ps4.executeUpdate();
               if(i > 0)
                  System.out.println("Record Deleted Successfully.");
               else
                  System.out.println("Employee Not Found.");
               break;

            // EXIT
            case 5:
               System.out.println("Program Terminated.");
               break;

            default:
               System.out.println("Invalid Choice.");
         }
      } while(choice != 5);
      con.close();

   } catch(Exception e) {
      System.out.println(e);
   }
```

```
    }
}
```

- Write a program of which insert a default record of an employee using a callable statement. (Procedure without parameters)
  - Open **phpMyAdmin**.
  - Select the database.
  - Click **Routines**
  - Click **Create Routine**
  - Enter:

```
Create this procedure provides name of the procedure
BEGIN
   INSERT INTO emp
   VALUES (101,'Hello','Manager','Rajkot',55000,'Sales');
END
```

```
import java.sql.*;

public class EmployeeInsertCallable {
   public static void main(String args[]) {

      try {

         // Load JDBC Driver
         Class.forName("com.mysql.cj.jdbc.Driver");

         // Create Connection
         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/test",
               "root",
               "");

         // Create CallableStatement
         CallableStatement cs = con.prepareCall("{call InsertEmployee()}");

         // Execute Procedure
         cs.execute();

         System.out.println("Record Inserted Successfully.");

         con.close();
```

```
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

- Write a program of which insert a record of an employee using a callable statement. (Procedure with parameters)

```
CREATE PROCEDURE InsertEmployee(
    IN p_empno INT,
    IN p_empnm VARCHAR(30),
    IN p_designation VARCHAR(30),
    IN p_city VARCHAR(30),
    IN p_salary DOUBLE,
    IN p_department VARCHAR(30)
)
BEGIN
    INSERT INTO emp
    VALUES(p_empno, p_empnm, p_designation, p_city, p_salary, p_department);
END;

import java.sql.*;

public class EmployeeCallable {

    public static void main(String args[]) {

        try {

            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Create Connection
            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/test",
                    "root",
                    "");

            // Create CallableStatement
            CallableStatement cs = con.prepareCall("{call InsertEmployee(?,?,?,?,?,?)}");
```

```
            // Set Parameter Values
            cs.setInt(1, 101);
            cs.setString(2, "Amit");
            cs.setString(3, "Manager");
            cs.setString(4, "Rajkot");
            cs.setDouble(5, 55000);
            cs.setString(6, "Sales");

            // Execute Procedure
            cs.execute();

            System.out.println("Record Inserted Successfully.");

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```